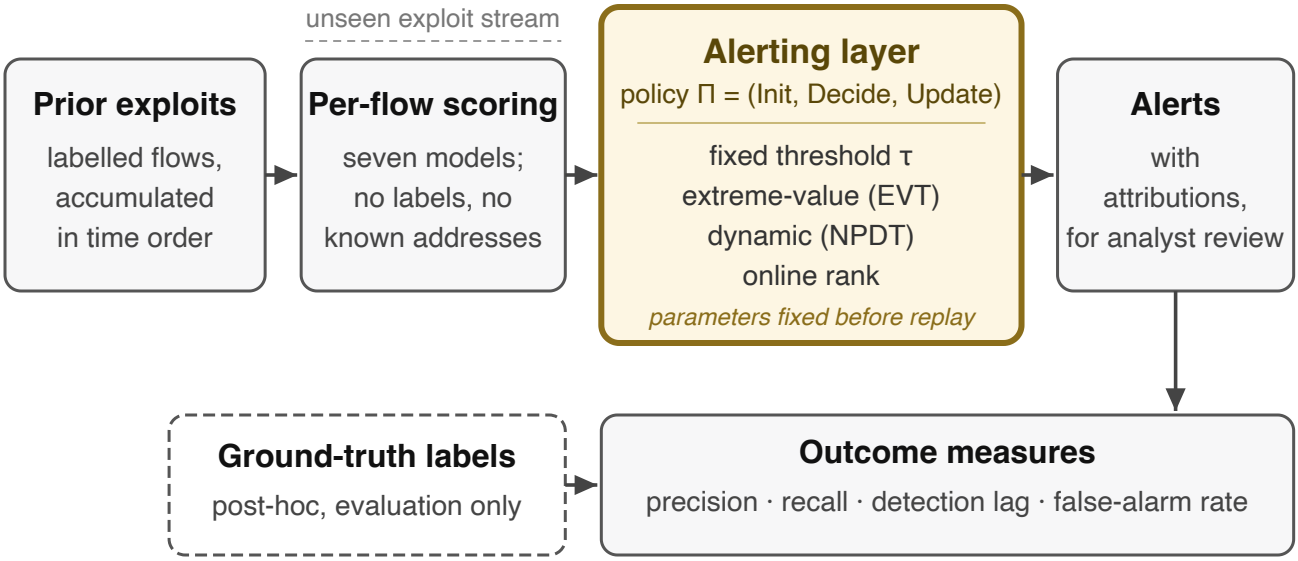
unseen exploit stream
Prior exploits
labelled flows, accumulated in time order
Per-flow scoring
seven models; no labels, no known addresses
Alerting layer
policy Π = (Init, Decide, Update)
fixed threshold τ
extreme-value (EVT)
dynamic (NPDT)
online rank
parameters fixed before replay
Alerts
with attributions, for analyst review
Ground-truth labels
post-hoc, evaluation only
Outcome measures
precision · recall · detection lag · false-alarm rate